Unit:
Introduction to Data Science and Big Data

Assignment title:
Google Local Research Team
Autumn 2024 – Summer 2025

Important notes

- Please refer to the *Assignment Presentation Requirements* for advice on how to set out your assignment. These can be found on the NCC Education website. Hover over 'About Us' on the main menu and then navigate to 'Policies and Procedures' then scroll to the 'Student Support' area.
- You **must** read the NCC Education document *Academic Misconduct Policy* and ensure that you acknowledge all the sources that you use in your work. These documents are available on the NCC Education website. Hover over 'About Us' on the main menu and then navigate to 'Policies and Procedures' then scroll to the 'Student Support' area.
- You **must** complete the *Statement and Confirmation of Own Work*. The form is available on the NCC Education website. Hover over 'About Us' on the main menu and then navigate to 'Policies and Procedures' then scroll to the 'Student Support' area.
- **Please make a note of the recommended word count as stated in the brief. The maximum word count limit is the recommended work count plus 10% leeway. Submissions which exceed the maximum word count limit will be marked only up to the maximum word count limit. Any text that exceeds the maximum word count limit will not be marked. You must follow the note on word count as stated in the Submission Requirements section of the brief.**
- You must submit a paper copy and digital copy (on disk or similarly acceptable medium). Media containing viruses, or media that cannot be run directly, will result in a fail grade being awarded for this assessment.
- All electronic media will be checked for plagiarism

Introduction

This assignment is compulsory as it is a major part of the formal assessment. Each student will work on datasets independently using variety of different data science techniques and visualisation methods. This will lead to the development of a method for processing the dataset inferred from the underlying trends that were identified during the visualisation stage. Each student will discuss in a report the reasons for choosing the appropriate methods throughout the data science lifecycle.

Scenario

You are a member of the Research and Development team of Google Local, i.e., a Google Map service that allows users to write reviews and add ratings of places they have visited. Your team has collected three large-scale datasets, namely 1) Users, 2) Reviews and 3) Places. Please see Google Local Folder that details the attributes and relationships of the three datasets. Note that the datasets contain real-world data which is publicly available.

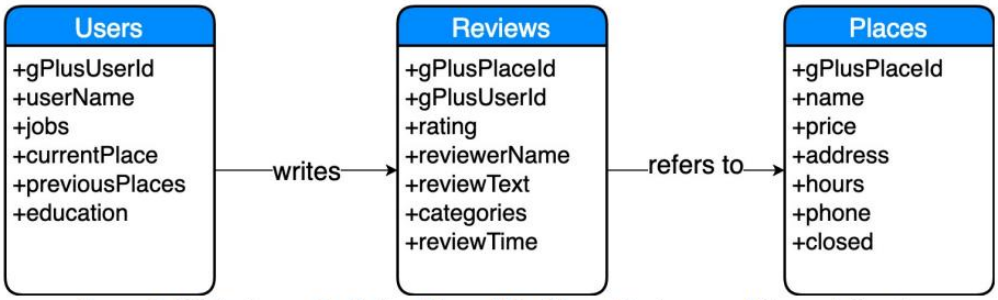


Figure 1. Attributes and relationships of the *Users*, *Reviews* and *Places* datasets.

Aim

Your job is to design, develop and evaluate an end-to-end system to efficiently store, process and analyse the three datasets. The system should support a range of different functionalities for storing, querying and automatically analysing the three datasets. The aim for each task in this assignment is for you to devise generic and custom strategies for the given problem for the company and advise them of any legal and ethical implications.

Task 1: Data Pre-processing – 25 Marks

The excel files provided contains approx. 20,000 location reviews. Perform the following tasks:

a) Prepare all the **THREE (3)** files for data processing. Perform the data cleaning such as convert the unstructured data to structured, spell check, remove contractions, tokenize the review column, etc. Explain the functions used to achieve the above task and the corresponding techniques in a report along with the evidence. You should provide a clear justification for the methods selected.

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Task 2: Data Processing – 25 Marks

In each question, it is important to discuss the technique used and provide justification and discuss results briefly.

a) Retrieve all places that have more than 3 reviews. e.g., output: #reviews place_id

5 Marks

b) Compute the number of places in the Places dataset that are permanently closed (the value of the 'closed' field in the Places dataset is 'True') and the number of places that are still operating (the value of the closed field is 'False'). e.g., output: #places is_closed.

5 Marks

c) Display the 20 most frequent phrases (i.e., sequence of 3 consecutive words) that appear in the review Text of reviews that have a rating of 5. e.g., Rating: 5.0. Output: I have been 351; I highly recommend 328.

5 Marks

d) Create a subset of the user's dataset by keeping only the 'IT Specialist' users (i.e., jobs attribute should be equal to 'IT Specialist').

5 Marks

e) Find the place (id and name) that has received the largest number of reviews. e.g., output: #reviews place_id name

5 Marks

Task 3: Data Analysis – 25 Marks

a) Display the number of positive, neutral and negative reviews using a suitable technique. Explain the step taken to achieve this task.

15 Marks

b) Justify your selected techniques in the above task and systematically compare them with other suitable techniques.

10 Marks

Task 4 – 25 Marks

a) Develop a strategy outlining how location owners can leverage the insights derived from Task 3 to enhance their business operations and customer satisfaction.

10 Marks

b) Google Local is planning to expand its data collection efforts to gather more detailed user preferences and behaviours for targeted advertising purposes. Evaluate the ethical considerations associated with this expansion and propose guidelines and safeguards to ensure responsible and ethical data collection practices and discuss the importance of ethical considerations in maintaining user trust and compliance with regulations such as General Data Protection Regulation (GDPR).

15 Marks

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Guidance

Consult your tutor if you are uncertain about any aspect of this assignment.

Submission requirements

You must submit a word-processed report.

Your report should answer Tasks 1 to 4. The word count for your report is **4000 words**.

Submission requirements

- Your submitted work must be word-processed, justified, and double-spaced.
- You will need to hand in a printed copy of your work, with the *Statement and Confirmation of Own Work* form attached.
- You will need to submit an electronic copy.
- You will need to include your name and NCC Education student number in the header on every page of your work.

Word Count

The word count for this assignment is **4,000 words**.

You **MUST** state the word count in the header of your assignment and declare the number of words on your assignment cover sheet. You **MUST** confirm that you have not exceeded the stipulated word limit by more than 10%.

Any words written beyond the stipulated limit will not be read by the examiners; no marks will be awarded, and no feedback provided for any text beyond the prescribed limit.

In accordance with accepted academic practice, the notion of word count includes the following without exception:

- All titles or headings that form part of the actual text. This does not include the fly (or cover) page or reference list.
- All words that form the actual work.
- All words forming the titles for figures, tables, and boxes. However, this does not include the text within boxes, tables, or figures themselves.
- All in-text (bracketed) references.
- All directly quoted material.

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Candidate checklist

Please use the following checklist to ensure that your work is ready for submission.

Have you read the NCC Education documents *What is Academic Misconduct? Guidance for Candidates* and *Avoiding Plagiarism and Collusion: Guidance for Candidates* and ensured that you have acknowledged all the sources that you have used in your work?	❑
Have you completed the *Statement and Confirmation of Own Work* form and attached it to your assignment? **You must do this.**	❑
Have you ensured that your work has not gone over or under the recommended word count by more than 10%?	❑
Have you ensured that your work does not contain viruses and can be run directly?	❑

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